

Kamaljit Aulakh

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EDUCATION

Syracuse University, College of Engineering and Computer Science, Syracuse, NY
Master of Science in Computer Science
Data Structures and Algorithms, Computer Security, Object Oriented Design

August 2023 – May 2025
GPA: 3.90

Mumbai University, Ramrao Adik Institute of Technology, Mumbai, India
Bachelor of Engineering in Information Technology
Web application development, Android Programming, Database Management Systems

August 2017 – June 2021
GPA: 3.74

EXPERIENCE

Software Engineer II, Vertex Inc - Pennsylvania, USA

June 2025 – Current

- Led the migration of legacy Jenkins CI/CD pipelines to GitHub Actions, implementing parallel and matrix test execution with PyTest, cutting test execution time by ~10%, enhancing reports and improving developer feedback latency across teams.
- Created an AI-driven workflow for Robot Framework failure analysis using Microsoft Copilot and Github copilot to extract root causes from CSV artifacts, reducing half a day of manual test triage to under two hours with a low-code/no-code solution.

Web and Data Applications Developer, Open Source Program Office - New York, USA

October 2024 – May 2025

- Built platform enabling 200+ faculty and students to explore **data storage solutions** using **WordPress**, integrating a custom theme and plugin using **PHP**, ensuring Web Content Accessibility compliance of UI using **WAVE**.
- Built a **JavaScript**-based **web application** to enable seamless data transfer between two research data repositories.

Global Integrations Team Summer Intern, Capri Holdings (Michael Kors) - New Jersey, USA

June 2024 – August 2024

- Designed an application using **Power Apps**, **Power Automate**, **SQL** and **Python** Script running on an EC2 instance to facilitate easy access to internal data records to monitor audit changes, and provide analytics via Power BI.
- Streamlined CDC process using **debezium SQL connector** to send data to **Kafka** topic enabling migration to an open source solution.

Software Engineer, IDFC FIRST Bank Ltd, Bangalore, India

June 2022 – August 2023

- Created RESTful services with **Java Micronaut** and **Python Flask** for mobile and internet banking for various features, resulting in a 10% increase in the opening of zero-balance savings accounts online.
- Led a team of 4 to build an admin panel implementing responsive UI components using **React.js** making it easier to render conversion data stored in NoSQL solution - **Aerospike DB**, generating reports, and diagrams plotted using the **React Google Charts** library.
- Implemented TDD with **JUnit** achieving 90% code coverage, integrated **SonarQube** for code quality, and used **JMeter** to optimize system performance.
- Used **Docker**, **Kubernetes**, and **Helm charts** to build and manage CI/CD pipelines on GoCD to achieve a 98% uptime in production deployments.

Systems Engineer, Tata Consultancy Services, Thane, India

Oct 2021 – June 2022

- Worked in an agile team writing backend logic across 10+ **microservices** using J2EE, **Spring MVC**, and **Spring Boot**, enhancing scalability and performance.
- Implemented front end features with **ReactJS**, **Redux** to integrate security enhancements (behavioral biometrics solution - BioCatch), leading to a 37% improvement in bot detection accuracy and reducing false positives by 20%.

PROJECTS

Cataract detection and classification: [Publication Link](#)

June 2020 – March 2021

- Teamed up with two classmates to conduct a comparative study identifying and classifying cataracts by comparing existing vectorization, machine learning, and deep learning models and subsequently building CNN model achieving an accuracy of 96% using combined feature vectors from the SIFT-GLCM algorithm
- **Built With:** Python, Keras, OpenCV, Flask, Matplotlib

SKILLS

Technical Skills: Java, Python, JavaScript, Docker, Kubernetes, Helm Charts, Data Structures and Algorithms, HTML5/CSS3, PHP, Aerospike, MongoDB, SQL, Unix/Linux, Automation/Manual software testing.

Tools: GoCD, Prometheus, Elasticsearch, Adobe XD, Jira, BitBucket, Jmeter, Grafana, Confluence, Postman, and Git.